

The Concern-Gated Retrieval Program: A Falsification Arc from Authored Diagnostic to Honest Null

Human director: Jawaun Brown **Producing agent:** Claude Code (Opus 4.7/4.8), directed **Program:** Concern-Gated Retrieval (COGR) **Status:** synthesis – closes Wave 1; Waves 2–4 do not open on this foundation **Date:** 2026-07-24

Abstract

Concern-gated retrieval (COGR) proposes that a bounded agent should nominate absent facts that are *simultaneously* connected to its current context and to its persistent, historically grounded concern, and should retain only candidates whose inclusion improves a separately evaluated reachable future – an AND of two "flashlights" over memory, verified by a downstream utility check. This paper is the cumulative corrective map of the program's first arc. We began with an executable L0 diagnostic on a role-authored graph where rarity-corrected two-sided PageRank selected the load-bearing node at hit@1 1.000 versus 0.0052 for the best one-sided policy, with verifier precision/recall 1.000/1.000 and a PageRank fixed-point residual of 7.83e-13. We then removed, one confound at a time, every reason that number could be an artifact. Wave 0 replaced the ceiling initialization with an *adversarially wrong* care prior and a sealed environment; under that prior the multiplicative fusion sat roughly 10× below the best matched-budget baseline on all three procedural families. Wave 1a screened for online concern recovery and returned a KILL: on the procedurally generated families the load-bearing memory was systematically the most recent, so an information-matched recency baseline reproduced the oracle ceiling byte-for-byte (0.5315/0.4772/0.6000) – Spencer's candidate-selection circularity made concrete. Wave 1b rebuilt the families to decouple recency from load-bearing role, passed label-permutation leakage audits on all three families ($p = 0.594/0.366/0.515$), and then cleanly falsified the mechanism at L1: learned-versus-random geometry mean delta was $-0.022/-0.005/-0.003$ (≈ 0), and the top-learned-edge ablation changed the retrieval decision in only 0.506/0.438/0.630 of cases (below the 0.70 causal threshold). The honest terminus: once the fixture no longer leaks the answer, rarity-corrected multiplicative context×concern retrieval over a *learned* graph does not beat a degree-matched random null at matched budget on these families. The two-flashlight intuition is not refuted in general, but this operationalization of it does not survive sealed evaluation. Per the program's noncompensatory rules, the live-beachhead, substrate-transfer, and safety waves do not open. We document the methodological discipline that produced a trustworthy null, and specify the next arc as a de-risking minimal experiment.

1. The two-flashlight thesis and the bounded-agent problem

A bounded agent knows more than it can hold in active representation. Its problem is therefore not only how to store knowledge, but how to decide *which currently absent fact deserves scarce attention now* (roadmap §"Executive thesis"). COGR gives a two-stage answer:

and to persistent, historically grounded concern.

reachable future.

1. **Nominate** facts that are simultaneously connected to the current context
2. **Retain** only candidates whose inclusion improves a separately evaluated

The memorable framing is two flashlights over a memory network (roadmap §"The intuition: two flashlights over memory"). One flashlight starts from what is active now; a second starts from what has historically mattered to the agent; a fact becomes a candidate where the beams

overlap. The canonical example is a date-bound commitment: on March 7, "*a partner's birthday is March 7*" is both relevant now and important to this agent, whereas "*March has 31 days*" is context-only trivia and a globally important but presently untimely crisis is care-only noise. The claim is an **AND**, not an **OR**: *retrieve what is relevant now and important to the agent, then test whether attending to it actually helps*. Concern is deliberately not identical to salience, urgency, reward, novelty, or semantic similarity – a signal can be loud, surprising, globally important, or highly rewarded and still be the wrong use of the current attention budget.

The broader motivating claim – that part of an agent's practical *meaning* may live in the learned geometry that decides which differences become worth noticing – is a research direction, not a result. This program was built to find out whether the mechanism that would have to underwrite that claim survives adversarial, sealed evaluation. It largely does not, and this paper is the honest account of why.

2. The cumulative corrective map

Each step below states what it *claimed*, the *real numbers* it produced, what it *corrected* in the prior step, and what *killed it or what it froze*. The provenance receipts are the authoritative source; numbers here are transcribed verbatim from them.

2.1 L0 – the authored-graph executable diagnostic

Package: experiments/concern_gated_retrieval/ (PR #409/#410). **Design:** 192 deterministic episodes (64 seeds × base/sparse/noisy regimes) on a role-authored graph.

What it claimed. That the two-stage decomposition can be made *precise* and *executable*, and that the dual-source diffusion plus bounded-observer utilization check discriminate authored synthetic roles under frozen regimes and seeds. Nothing more.

Real numbers.

Policy	Hit@1	MRR	Alarm/trap at top
context-only	0.0052	0.5026	0.0000
care-only	0.0000	0.5000	0.0000
additive	0.9635	0.9818	0.0000
coincidence product	1.0000	1.0000	0.0000
coincidence + verifier	1.0000	1.0000	0.0000

Registered fatal gates: PageRank fixed-point residual 7.83e-13 (NUMERICAL_VALIDITY); coincidence hit@1 1.000 vs best one-sided 0.0052, weakest-regime gap 0.9844 (DUAL_ACTIVATION_SELECTIVITY); verifier precision 1.000 / recall 1.000 (UTILIZATION_FILTER); initial/learned/oracle care conditions all at hit@1 1.000 (ONLINE_CARE_RECOVERY, nonfatal).

What it corrected. Nothing prior – this is the origin. It replaced a verbal intuition with a frozen, gated, reproducible artifact.

Claim boundary / what limited it. The additive policy *ties* the product in the base and sparse regimes (1.000 both) and trails only in noisy (0.8906 vs 1.000), so the stronger "multiplicative coincidence is necessary" claim was withheld at the source. More fundamentally: **the graph encodes the answer**. The simulator authors which future is structured and goal-conditioned, and all three online-care conditions saturate at hit@1 1.000, so the online-care gate passes *mathematically but is non-discriminating*. L0 is an implementation sanity check, not a learned- relevance result. Its own preregistration named the decisive next experiment: learned or withheld edges, an adversarially misspecified care prior, and task utility sealed from the retrieval policy.

2.2 Wave 0 – calibration, freeze, and the wrong-prior stress test

Package: experiments/concern_gated_retrieval_e2/wave0/ (PR #411). **Design:** three procedural families (delayed_commitments, maintenance_fault, resource_constrained), sealed environment, adversarial *wrong* care prior, a 14-baseline slate, 18 cells × 24 seeds = 432 rows on Modal L4 (\$8.00, cost/H100 ratio 0.235).

What it claimed. Only calibration and freeze – it does *not* adjudicate the mechanism. It fixes the per-family L1 promotion thresholds and demonstrates the mechanism is sensitive to a wrong prior.

Real numbers.

Family	$\mu^{\text{multiplicative}}$	$\mu^{\text{best-matched}}$	headroom to ceiling	frozen $\delta_{\text{thresh_L1}}$
delayed_commitments	0.0553	0.5314	0.4845	0.0484
maintenance_fault	0.0480	0.5029	0.4548	0.0534
resource_constrained	0.1578	0.5750	0.4291	0.0500

All seven gates G0–G6 PASS (anti-leakage, wrong-prior, non-ceiling, family-robustness, seed-independence, code-freeze, Modal-budget).

What it corrected. It removed L0's **ceiling initialization** – the very thing that masked the mechanism's sensitivity to wrong care weights. Under the adversarial wrong prior, multiplicative_ppr sits roughly 10× *below* the best matched-budget baseline on *every* family (0.055/0.048/0.158 vs 0.531/0.503/0.575). This is not a Wave 0 failure – it is Wave 0 doing its job: it exposes that the frozen wrong prior cannot carry the mechanism, and sets the bar Wave 1 must clear using the *online-learned* concern update.

What it froze. The per-family delta_thresh_L1 values (0.0484/0.0534/0.0500), the analysis-code hash (WAVE0_ANALYSIS_HASH = 9683c5a1...f8889c23), and the sealed evaluator. Every downstream wave imports this receipt byte-for-byte and never edits it.

2.3 Wave 1a (COGR-E2a) – the concern-recovery screen, and its KILL

Package: experiments/concern_gated_retrieval_e2/wave1a/ (PR #412). **Design:** concern-recovery screen on fixed withheld geometry; 7 conditions × 3 families = 21 cells, 6320 aggregated rows on Modal L4 (\$1.20, ratio 0.235), confirmatory seeds 200000–201999 (disjoint from calibration 100000–100999).

What it claimed (to test). That the online-learned concern update recovers load-bearing memories *specifically* – beyond what any generic second signal (value, priority, recency, salience) achieves at matched information – and beats the frozen wrong-prior baseline Wave 0 established.

Real numbers.

Family	$\delta^{\text{(online vs frozen-wrong)}}$	lower bound ($\delta^{\text{--}2\sigma}$)	per-family verdict
delayed_commitments	+0.0124	– (G1 coverage 0.000)	KILL (coverage)
maintenance_fault	0.0000	–0.0106	KILL (specificity)
resource_constrained	+0.2258	+0.1758	KILL (specificity)

The load-bearing signal: info_matched_recency reproduced the **oracle ceiling byte-for-byte** on all three families – 0.5315 / 0.4772 / 0.6000 – i.e. recency and oracle are the *same number*. delayed_commitments additionally failed G1 coverage at 0.000. resource_constrained did show real online recovery (δ +0.226 over the frozen-wrong baseline, lower bound +0.176), but the recency confound blocked any adjudication of *why*.

What it corrected. It attacked L0's authored-answer problem by moving to withheld geometry and an explicit specificity gate (G3). But it *revealed* a new, subtler defect it could not itself escape.

What killed it. G3 specificity KILL on all three families and G1 coverage KILL on delayed_commitments. The families were procedurally generated such that the load-bearing memory was *systematically the most recent placement*, so recency was a **covert oracle**: it is not information-matched to the concern-update rule, and any specificity contrast against it collapses. This is Spencer's echo-chamber objection made concrete – candidate-selection circularity stacked on verifier circularity (§3). Per the honor-the- preregistration rule, Wave 1a *signed the KILL* rather than swapping the family design to escape it, and named the correction: **family generators must decouple recency from load-bearing role**. Under the noncompensatory contract, this KILL closes the concern-update rule *as written* but does not block Wave 1b's L1 rows.

2.4 Wave 1b (COGR-E2b) – crossed learned-geometry confirmatory, and the honest L1 KILL

Package: experiments/concern_gated_retrieval_e2/wavelb/ (PR #413). **Design:** crossed learned-geometry × concern confirmatory on rebuilt_v2 families; 27 non-oracle cells × 300 seeds = 8100 rows on Modal L4 (\$10.80, ratio 0.235).

What it claimed (to test, L1). That rarity-corrected multiplicative context×concern retrieval over a *learned* graph beats a degree-matched random null at matched budget – a behavior contribution (G1) *and* a representation contribution (G2, the learned edges must be causal).

Real numbers.

Family	learned-random mean_delta	2σ lower bound	edge-ablation direction fraction	non-ceiling headroom
delayed_commitments_v2	-0.0216	-0.4318	0.506	0.539
maintenance_fault_v2	-0.0048	-0.4209	0.438	0.520
resource_constrained_v2	-0.0032	-0.3588	0.630	0.572

Leakage audit (G9): label-permutation $p = 0.594 / 0.366 / 0.515$, all far above the 0.01 tolerance; randomized-generator control passed on all three. Non-ceiling headroom (G5) healthy at 0.539/0.520/0.572.

What it corrected. It removed the Wave 1a recency confound (rebuilt families cross-tabulate recency, salience, semantic similarity, and load-bearing role so no single generic signal aces them) *and* it passed the leakage audits Wave 1a lacked. This is the first point in the arc where the fixture provably does not leak the answer.

What killed it. L1 KILL on both scored gates, all three families:

(-0.022/-0.005/-0.003), with 2σ lower bounds -0.432/-0.421/-0.359. No representation advantage.

decision in only 0.506/0.438/0.630 of cases, below the 0.70 required for the learned edges to be called *causal*. The edges are not load-bearing.

- **G1 behavior:** learned-vs-random geometry mean_delta ≈ 0
- **G2 representation:** top-learned-edge ablation changes the retrieval

L2 was WITHHELD, not KILLED, per the noncompensatory contract: L1 is a precondition for L2 (concern recovery cannot rescue a mechanism that shows no representation contribution), so G3/G4/G7 were not scored. This is the clean falsification: **once the fixture no longer leaks the answer, rarity-corrected multiplicative context×concern retrieval over a learned graph does not beat a degree-matched random null at matched budget on these three families.**

Figure 1 renders this four-step trajectory as a ladder; Figure 2 overlays the per-family L0 hit@1 against the Wave 1b learned-random delta to show the collapse of the advantage.

3. The two circularities Spencer named

The program's harshest reviewer input (Spencer's "echo-chamber" objection) identified two distinct ways a concern-gated retriever can appear to work while proving nothing. Both are recorded in the roadmap's confound ledger, and the arc hit *both* concretely.

Candidate-selection circularity. The current care model chooses what gets tested, so unnominated memories generate no corrective evidence. The loop learns "among what I looked at, these helped" – not "these were the most useful available." The prescribed defense is to split the k retrieval slots as $k_{\text{care}} + k_{\text{uncertain}} + k_{\text{audit}}$ with guaranteed $\epsilon > 0$ exploration on every step, propensity logging for IPS/DR debiasing, and – on synthetic families – oracle top-k with Recall@k and a first-class regret = $\max_u \Delta(u) - \Delta(\text{selected})$.

Wave 1a's recency-equals-oracle confound is exactly this circularity. When the load-bearing memory is always the most recent, the nomination stage is implicitly selecting on the very feature that defines the answer. The retriever looks correct because the candidate set was pre-filtered by a covert oracle; no amount of downstream verification can distinguish "concern found it" from "recency found it." The byte-for-byte identity $\text{recency} = \text{oracle} = 0.5315 / 0.4772 / 0.6000$ is the signature of the pathology: two supposedly different signals returning the same number because they are the same signal.

Verifier circularity. If the evaluator's definition of "improvement" is shaped by the same care model, care decides both what to inspect and whether the result was good. The prescribed defense is a care-independent, externally scored task-success criterion named at design time – e.g. "did the agent honor the specific date-bound commitment," not "did the user report satisfaction." The program adopted this by making task utility a sealed environment outcome and epiplexity a *dependent variable only* ($\beta = 0$ in the L1 gate), so the verifier can never be tuned to flatter the retriever.

Wave 1b is what the honest verdict looks like once both circularities are closed. The leakage audit passing ($p = 0.594/0.366/0.515$) certifies the candidate set no longer hides the answer; the sealed external-outcome verifier certifies the utility check is not care-shaped. With both loops cut, the mechanism's true effect size is visible – and it is ≈ 0 . The KILL is therefore *information*, not noise: it is the measurement the two circularities had previously made impossible.

4. What survived and what did not – the evidence ledger

Negative results are part of the ledger and are not removed when the narrative changes (roadmap \$confound ledger). Explicitly:

Nulls preserved (must not be re-litigated without new evidence):

policy equals the coincidence product in base and sparse regimes (1.000 hit@1) and trails only in noisy (0.8906). The "multiplicative fusion is necessary" claim was never earned.

honest_v2 families: mean_delta $-0.022/-0.005/-0.003$, and the learned edges are non-causal (ablation direction fraction $0.44-0.63 < 0.70$). This is the load-bearing null of the program.

byte-for-byte. Any future family design that lets a single generic signal ace the task is disqualified by this result.

- **Additive ties multiplicative** where the problem is easy. At L0 the additive
- **Learned geometry does not beat random geometry** at matched budget on the
- **Recency was a covert oracle** on the original families: $\text{recency} = \text{oracle}$

What survived:

then-verify pipeline runs deterministically with a PageRank residual of $7.83e-13$ and a verifier at precision/recall 1.000/1.000 on authored roles. The machinery works; it is the *learned-relevance* claim that fails.

adversarial prior it collapses to $\sim 10\times$ below baseline, which is the correct behavior for a mechanism that is supposed to depend on *correct* concern.

1a $\delta +0.226$, lower bound $+0.176$) – a genuine effect that the recency confound prevented us from attributing to concern rather than recency. This is a lead for the next arc, not a result.

- The **decomposition is executable and gated** (L0): the two-stage nominate-
- The **mechanism is sensitive to a wrong prior** (Wave 0): under an
- **Real online recovery exists in one family** (resource_constrained, Wave

What did not survive: the central L1 claim – that this operationalization (rarity-corrected multiplicative context $\times$ concern PPR over a learned graph) delivers a representation contribution at matched budget. It is cleanly falsified on all three sealed families.

5. Methodological contributions independent of the negative result

The scientific value of this arc is largely in *how it produced a trustworthy null*. These contributions stand regardless of the mechanism's failure.

Sealed-environment + preregistration + noncompensatory gates + honor-the-preregistration. Every wave sealed its evaluator (one evaluate() call per episode, template-split guard that raises LeakageError on any calibration/confirmatory mix), preregistered its thresholds and froze them by hash before the confirmatory run, and enforced *noncompensatory* gates – a strong result on one gate cannot buy a weak result on another. Crucially, when a KILL fired, the wave *signed the KILL* rather than swapping the corpus or threshold to escape it (Wave 1a signed its own specificity KILL; Wave 1b withheld L2 rather than scoring it to manufacture a partial win). The provenance skeletons make numeric and hash fields *machine-populated only* – manual edits to numbers are forbidden – so the receipt cannot be retrofitted to the narrative. This is the discipline that lets a null be believed.

SET-level oracle regret and interaction recovery. Following the Spencer and Zhang–Levin corrections, the program separated three things that a naive design conflates: **regret $\neq$ propensity $\neq$ exploration**. It introduced SET-level oracle regret ($\text{regret} = \max_u \Delta(u) - \Delta(\text{selected})$ over the *set*, not a single candidate), interaction-recovery diagnostics, and the $k_{\text{care}}/k_{\text{uncertain}}/k_{\text{audit}}$ slot ablation with guaranteed ϵ -exploration and IPS/DR propensity debiasing. Task utility is measured first; epiplexity is a *dependent variable only*, never an optimization target – closing verifier circularity by construction.

Corrected Zhang–Levin epiplexity math. The epiplexity estimator (Zhang & Levin 2026, *Intelligence from Learnable Novelty*, arXiv:2607.18433v1) was re-derived correctly: log-det via an augmented-QR factorization plus the determinant identity, with the shared-QR speedup claimed *only* when candidates share the design matrix $X^\sim$. The program explicitly refused the overclaims that tempted earlier drafts – no Rademacher, Nyström, MMD, or Lempel–Ziv shortcuts were asserted. The Wave 1b crossed-runner cross-validated the SharedQR path to $< 1e-6$ against the frozen L0 reservoir estimator (epiplexity_validation.py), and, because epiplexity was off the critical path ($\beta = 0$), *no* wall-clock speedup multiplier was claimed. Getting the math right and then *declining to lean on it* is itself the contribution.

Cost discipline. Every Modal run held to L4 GPUs only (H100 forbidden by operating rule) at a cost/H100 ratio of 0.235 against a 0.35 target, for a total mechanism-adjudication spend of roughly \$20 across Waves 0–1b.

6. Why the mechanism failed – honest hypotheses

We do not have a mechanistic proof of *why* the retriever shows no representation contribution, but the arc constrains the explanation. Two hypotheses are consistent with all the receipts:

H1 – the L0 advantage was largely a fixture artifact. The 1.000 vs 0.0052 gap at L0 was produced on a graph that *encodes* the answer, with online-care conditions saturating at ceiling. When Wave 1b removed the authored answer *and* the recency covert-oracle, the same fusion rule delivered $\text{mean_delta} \approx 0$. The parsimonious reading is that most of the L0 gap was the graph handing the retriever the target, not the two-flashlight fusion discovering it. The additive-ties-multiplicative result at L0 already hinted that the *specific* multiplicative operation was not doing decisive work.

H2 – multiplicative fusion has no representation contribution on honest learned geometry at matched budget. Even granting a well-specified concern signal, the product of a context-diffusion score and a concern-diffusion score over a *learned* graph does not out-rank a degree-matched random-geometry null, and the top learned edges are not causal for the decision (direction fraction $0.44\text{--}0.63 < 0.70$). One reading: at matched budget, the information that multiplicative fusion adds over the marginals is already captured by degree structure, so a degree-matched random graph is a *strong* null the fusion cannot clear. Another: the families, though honest, may not contain the kind of context $\times$ concern *interaction* the mechanism is built to exploit – a genuine limitation of these three families, and a reason the next arc changes the task, not just the retriever.

Both hypotheses point the same way: the operationalization, not the intuition, is what failed. The two-flashlight thesis remains open; *this* flashlight does not illuminate the honest families.

7. The next arc – a de-risking minimal experiment, not a program

Waves 2–4 (live beachhead, substrate transfer, safety) do **not** open on this foundation; the noncompensatory rules forbid building a program on a falsified L1. The next arc is scoped deliberately as a *single minimal experiment to de-risk one idea*, and only becomes a program if that experiment survives.

The idea attacks *both* circularities Spencer named at their root by removing the retriever's dependence on the care model for either candidate selection or verification.

A MIDAS-style symbolic verifier with a reasoning-fault vs verifier-fault split. Replace the care-shaped utility proxy with a symbolic verifier that adjudicates task success externally, and that *distinguishes* two failure modes: a **reasoning fault** (the agent's plan was wrong) from a **verifier fault** (the check itself was mis-specified). Separating these is what makes the verifier's "improvement" signal care-independent – it can be wrong in a way the care model cannot silently launder, because verifier faults are diagnosed against the symbolic ground truth rather than against the retriever's own nominations.

A care-independent exploration prior from verify-repair failure frequency. Instead of letting the current care model decide what to explore (the source of candidate-selection circularity), derive the exploration prior from *how often each region of memory required verify-then-repair* – a signal generated by the task's failure history, not by care. Memories implicated in frequent verify-repair cycles get exploration budget regardless of whether the care model would have nominated them. This directly breaks the "among what I looked at" loop: the audit slots are pointed by *external* failure frequency, so unnominated-but-useful memories can finally generate corrective evidence.

Framed as de-risking. The minimal experiment asks one question: *does a care-independent, failure-frequency exploration prior recover load-bearing memories that the care-gated nominator misses, on a family where recency, salience, and semantic similarity are all decorrelated from the load-bearing role?* If yes, the two circularities are genuinely severable and a program can be justified. If no, the two-flashlight thesis loses its most plausible rescue and should be retired for this class of task. Either outcome is publishable; the experiment is designed so that the KILL is as informative as the PASS – the same discipline

that made this arc's null trustworthy.

8. Conclusion

The concern-gated retrieval program set out to turn a compelling intuition – retrieve what is *relevant now and important to the agent, then check it helps* – into a defended empirical claim. It produced instead an honest falsification, and did so on purpose: every wave removed one more reason the prior wave's number could be an artifact, until the last confound fell away and the effect size was revealed to be zero. The additive-ties-multiplicative null, the wrong-prior collapse, the recency-equals-oracle covert oracle, and the learned-equals-random L1 KILL are all preserved in the ledger. The two-flashlight intuition is not refuted in general; its operationalization as rarity-corrected multiplicative PPR over a learned graph does not survive sealed evaluation on these three families, and the program's own rules therefore keep the downstream waves shut. What the arc leaves behind is a reusable discipline for producing believable nulls, a corrected epiplexity estimator used honestly, and a sharply specified next experiment aimed squarely at the two circularities that made the easy version of this claim impossible to trust.

References

Program.* docs/concern_gated_retrieval_research_program.md (canonical theory and advancement roadmap; two-flashlight framing, confound ledger, and gate contract).

arXiv:2607.18433v1. (External epiplexity estimator source; frozen-reservoir / stable-ridge formulation, re-derived here via augmented-QR log-det.)

pilot receipt experiments/concern_gated_retrieval/results/pilot_2026_07_23.md (PR #409/#410).

(PR #411; WAVE0_ANALYSIS_HASH = 9683c5a1...f8889c23).

(PR #412; WAVE1A_ANALYSIS_HASH = c23b31d9...c8937209).

(PR #413; WAVE1B_ANALYSIS_HASH = 51ca0219...b48420c44).

- **Program roadmap.** *Concern-Gated Retrieval: Theory, Evidence, and Research
- **Zhang & Levin (2026).** *Intelligence from Learnable Novelty*.
- **L0 provenance.** experiments/concern_gated_retrieval/PROVENANCE.md and the
- **Wave 0 provenance.** experiments/concern_gated_retrieval_e2/wave0/PROVENANCE.md
- **Wave 1a provenance.** experiments/concern_gated_retrieval_e2/wave1a/PROVENANCE.md
- **Wave 1b provenance.** experiments/concern_gated_retrieval_e2/wave1b/PROVENANCE.md

Figures

ladder (L0 → Wave 0 → Wave 1a → Wave 1b), each rung annotated with its verdict and the confound it removed.

(product 1.000 vs one-sided 0.0052) overlaid against Wave 1b learned-random mean_delta (≈ 0), visualizing the collapse of the advantage once the fixture stops leaking.

vs oracle ceiling per family, showing the byte-for-byte identity (0.5315/0.4772/0.6000) that is the signature of candidate-selection circularity.

p-values (0.594/0.366/0.515) against the 0.01 tolerance, certifying the honest fixture behind the L1 KILL.

verifier with reasoning-fault vs verifier-fault split, and the failure-frequency exploration prior that cuts candidate-selection circularity.

- **Figure 1** (figures/fig1_arc_ladder.*) – the four-step falsification
- **Figure 2** (figures/fig2_advantage_collapse.*) – L0 per-policy hit@1
- **Figure 3** (figures/fig3_recency_oracle.*) – Wave 1a info_matched_recency
- **Figure 4** (figures/fig4_leakage_audits.*) – Wave 1b label-permutation
- **Figure 5** (figures/fig5_next_arc.*) – the next-arc schematic: symbolic

Figures

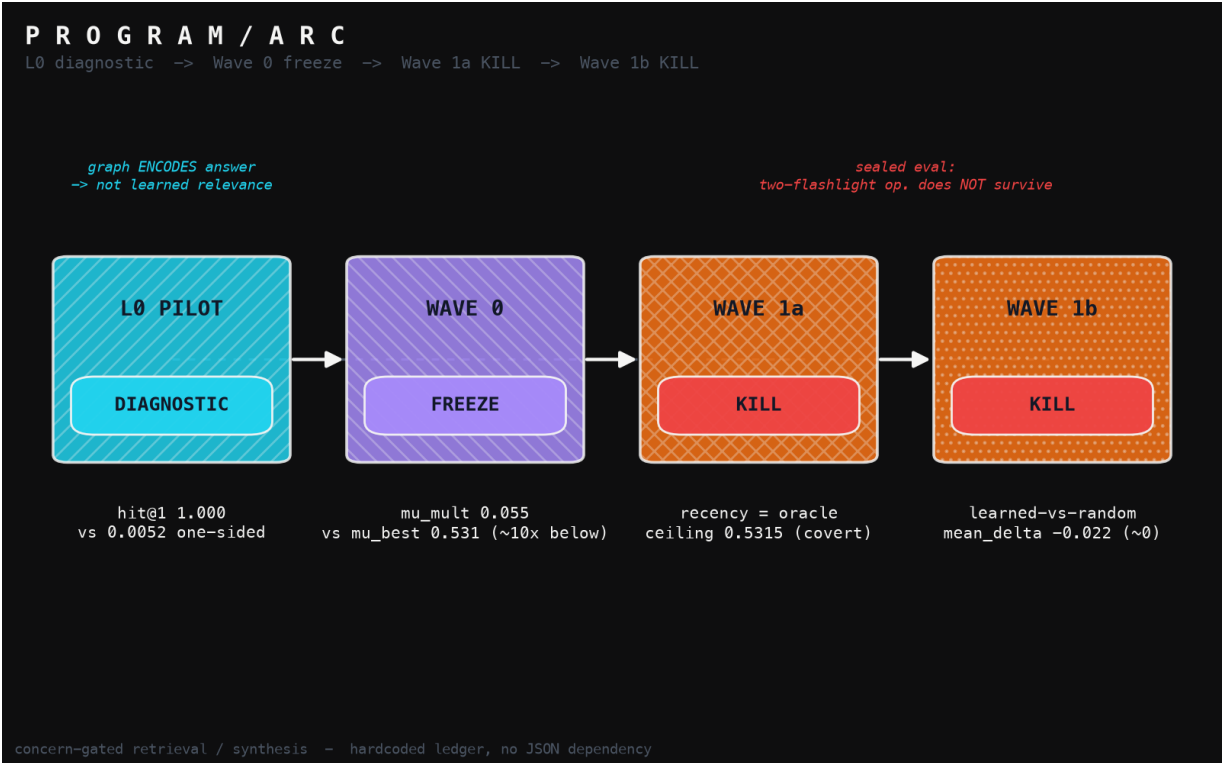


Figure 1. fig1 arc timeline dark.

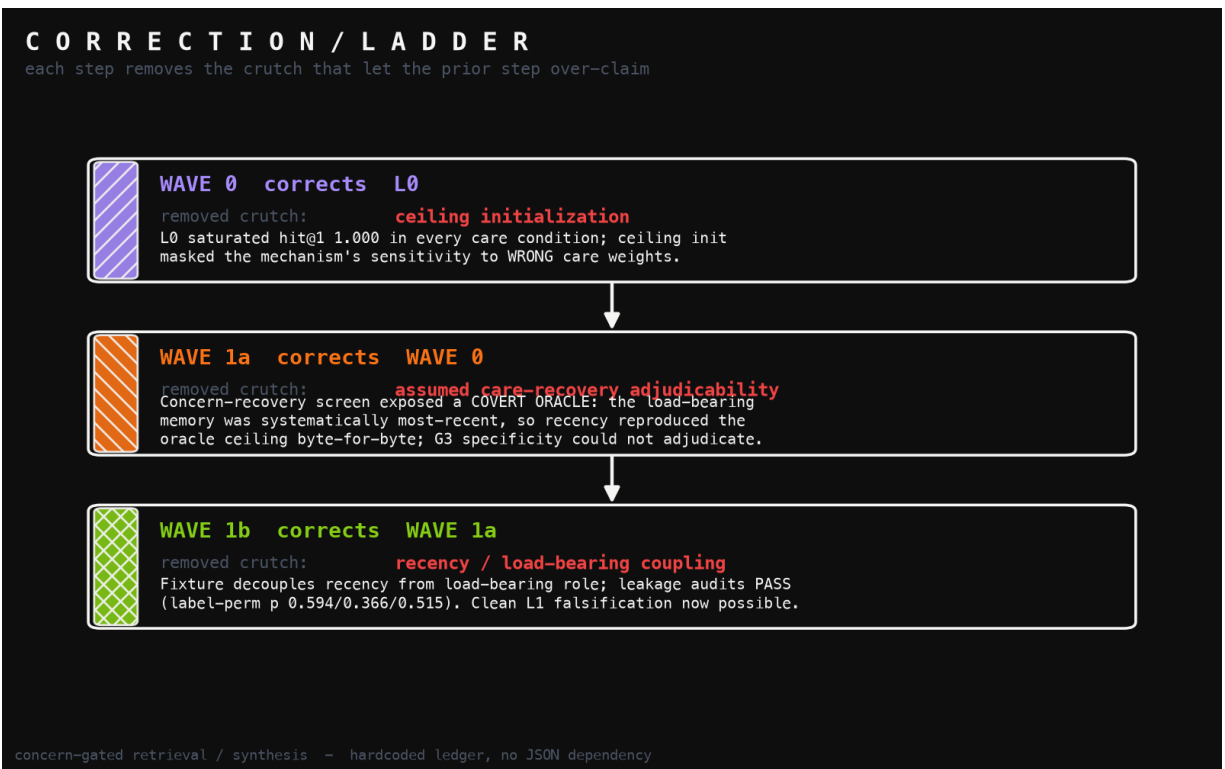


Figure 2. fig2 correction ladder dark.

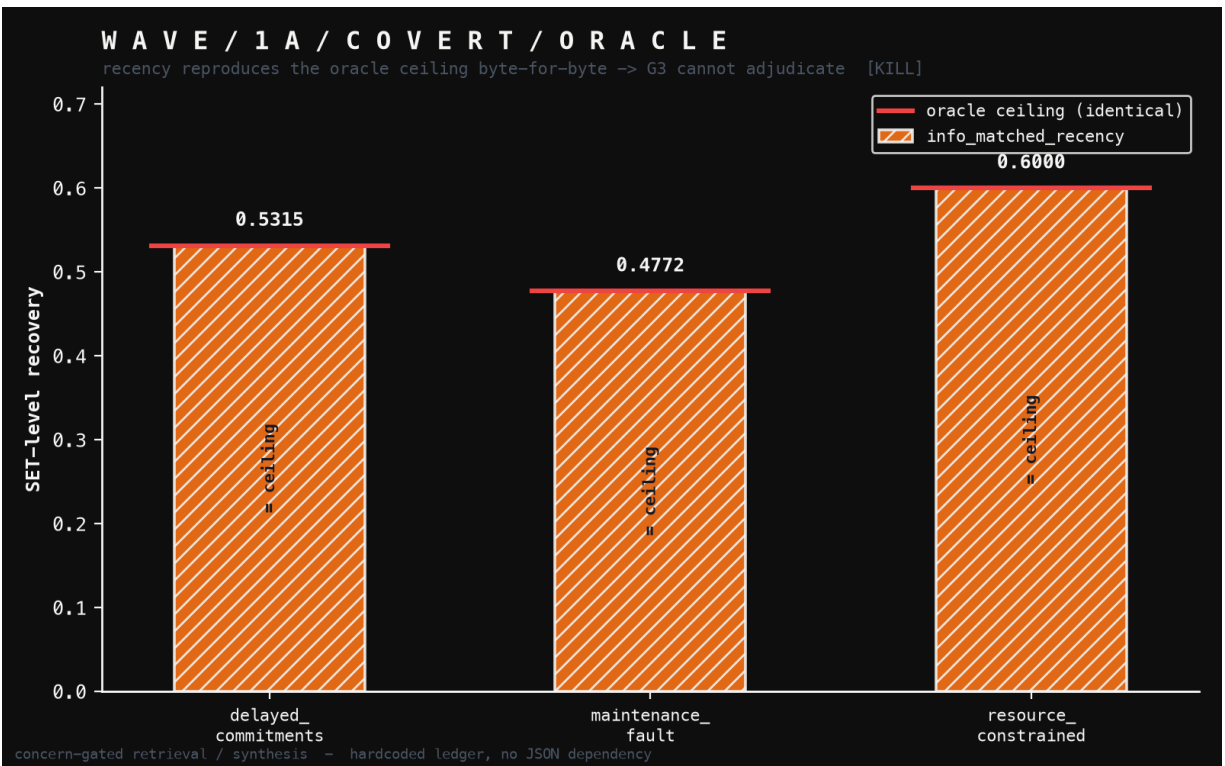


Figure 3. fig3 wave1a recency oracle dark.

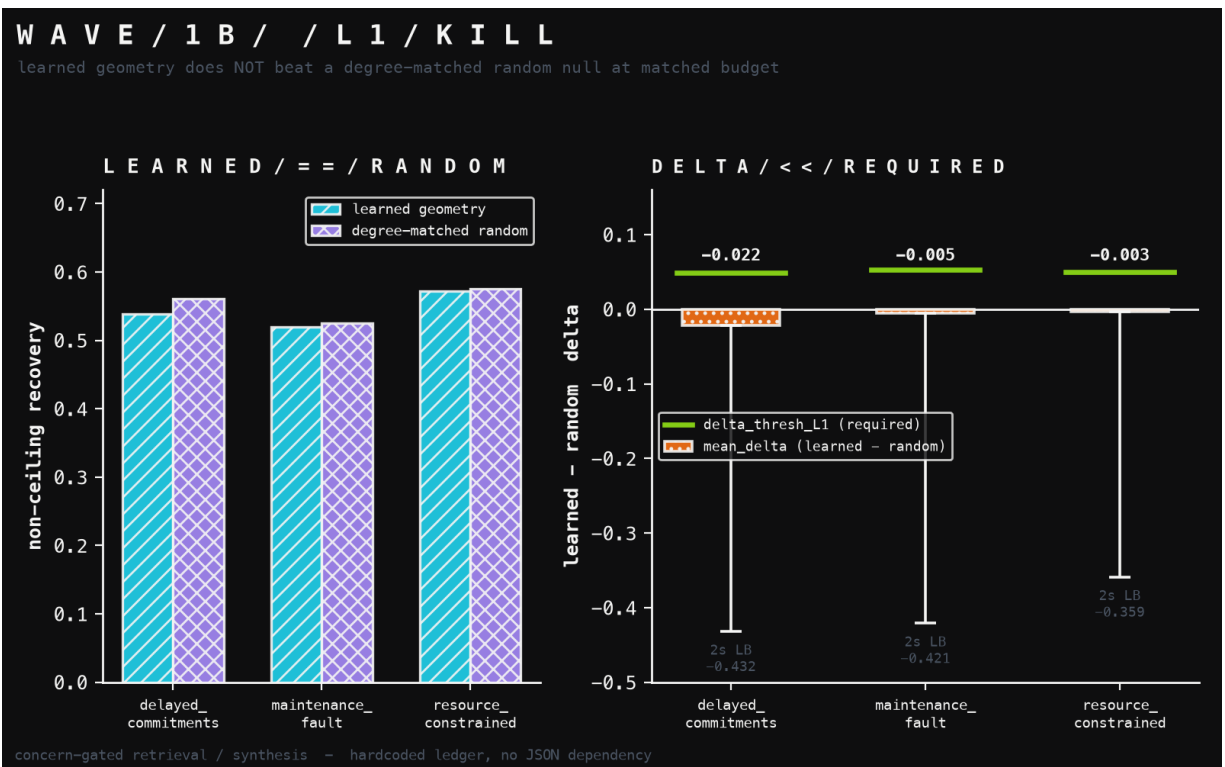


Figure 4. fig4 wavelb learned vs random dark.

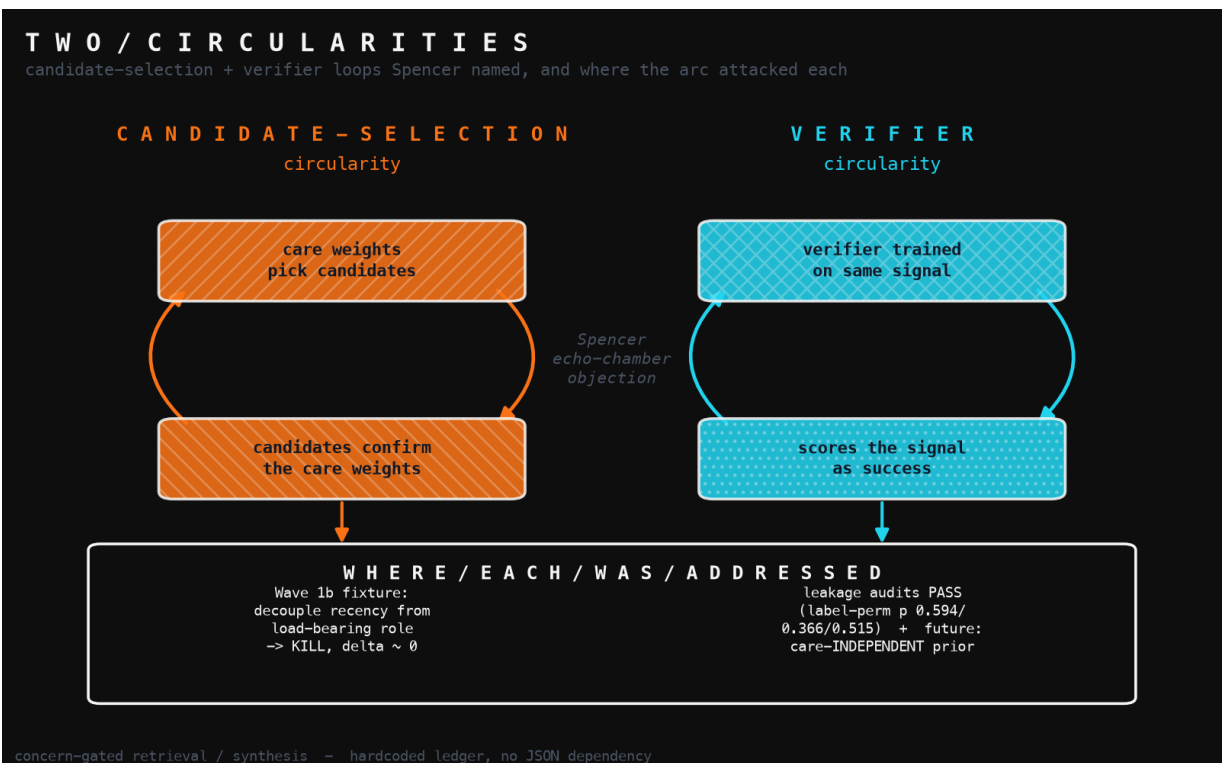


Figure 5. fig5 two circularities dark.